NOISEMAP · PRODUCT REQUIREMENTS AND SPECIFICATIONS

NoiseMap product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-02	Amish Chadha	Kit 1.8 migration from NSM-REQ-001 to PRS; each R becomes a PR with the same target, verification and status, traced to a UR (NSM-URS-001); tolerance and units added; PR-016 and PR-017 added
0.5	2026-10-01	Amish Chadha	Status figures from NSM-CAL-001 v0.3 (constructable design, NSM-DDR-003); R13 stated against the value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3; R13 redefined to the NoiseMap parts under NSM-DDR-001 D1; R2 metric set adopted (D5); status column from NSM-CAL-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (numbered, measurable requirements with targets and concept status)
0.1	2026-09-25	Amish Chadha	Initial scaffold (as NSM-REQ-001)

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What the design must do, each requirement traced to a user need in NSM-URS-001. This document replaces NSM-REQ-001: each R number is now a PR number (old number in the "Was" column), with the same target, verification and status. Targets are proposals for review with a co-design partner and must be revised with one before the design is frozen. The metric set (PR-002) and the budget scope (PR-013) follow NSM-DDR-001 D5 and D1; the relaxed mass limit (PR-010), the wind flag method (PR-011) and the interval rule (PR-012) follow NSM-DDR-002; all were decided by Amish, 2026-09-25: go with recommendation. Status comes from the TRL 3 calculation note NSM-CAL-001 v0.3; "met on paper" means met by calculation only. PR-016 and PR-017 are new in this migration and describe parts of the design that were already in the build plan. Seventeen requirements: one not met on paper (PR-003), two at risk (PR-004, PR-005), six to verify at TRL 4 (PR-008, PR-009, PR-011, PR-014, PR-016 and the install time of PR-010), the rest met on paper or by design.

Design case: one node on a 114.3 mm street pole 0.45 m behind the curb, microphone 4.0 m above the sidewalk and 0.45 m from the pole face, in a mixed street with traffic by day and bars at night. This is concept option A of NSM-OPT-001; options B and C are scored against the same user requirements and are not designed to these targets.

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Software	Levels only	No audio samples are stored or leave the microphone head; only levels cross the cable or the radio.	No data storage in the head; cable carries levels only; firmware published with a build hash and read-out protection	none	n/a	Design review, firmware audit, cable capture (TRL 3 design review; audit and capture at TRL 4)	Met by design; the guarantee rests on the firmware, since the cable could carry a speech codec (NSM-DDR-001 D2)	R1
PR-002	UR-002	Software	Reported metrics	Each 15 minute record (at SF7 to SF9; at SF10 to SF12 the interval and the fifteen sub-interval values lengthen, 1.6 to 5.8 minutes each, NSM-DDR-002) shall carry fifteen 1-minute LAeq, LAeq, LAFmax, LAFmin, L10, L90, LCeq and a device status.	22 bytes per record	exact	bytes	Design review (TRL 3)	Met by design at SF7 to SF9; sub-intervals longer at SF10 to SF12	R2
PR-003	UR-003	Performance	Accuracy	LAeq,15min shall agree with a co-located class 1 meter after field calibration.	2	max, 40 to 100 dBA	dB	Analysis (NSM-CAL-001 C3, TRL 3); side-by-side field test (TRL 4)	Not met on paper: expanded uncertainty 2.9 dB (3.1 dB with the reference meter) on assumed terms	R3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									against 2 dB; facade bias up to 2.4 dB (2.7 dB in a canyon) before a site correction; relaxing the target is Amish's decision	
PR-004	UR-004	Performance	Measuring range	The node shall measure from 35 to 110 dBA (A-weighted).	35 to 110	min and max	dBA	Analysis (NSM-CAL-001 A1, TRL 3); bench test (TRL 4)	At risk: ICS-43434 34.9 dBA to 113 dB SPL, which is 103 to 108 dBA for traffic; IM72D128 (fitted by default) 27.9 dBA to 119 dB SPL, 109 to 114 dBA	R4
PR-005	UR-005	Performance	Frequency weighting	A and C weighting shall follow IEC 61672-1 within class 2 tolerances from 63 Hz to 8 kHz (goal).	Class 2, 63 Hz to 8 kHz	class 2 limits	Hz	Analysis (NSM-CAL-001 B, TRL 3); bench test (TRL 4)	At risk: roll-off equalized to 0.8 dB at 63 Hz; membrane, windscreen and head diffraction unknown	R5
PR-006	UR-006	Software	Time stamps	Each record shall be time-stamped within 2 s of UTC.	2	max	s	Design review (LoRaWAN network time, TRL 3)	Met by design	R6
PR-007	UR-007	Electrical	Energy	The node shall be energy	1.5 h; 14 days	min	h; days	Analysis (NSM-CAL-001 D	Met on paper (5.81 Wh/day for	R7

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				neutral at 1.5 peak sun hours and run 14 days without sun.				and E, TRL 3); bench test (TRL 4)	0.29 Wh/ day; 53 days)	
PR-008	UR-010	Materials	Ingress and weather	The enclosure shall be IP65 and the microphone head shall survive driving rain from -20 to +50 degrees C.	IP65; -20 to +50	range	IP code; degrees C	Spray test and thermal chamber (TRL 4)	To verify at TRL 4	R8
PR-009	UR-012	Usability	Calibration check	A field check with a 94 dB, 1 kHz calibrator shall take 10 minutes or less and store the offset on the node.	10	max	min	Timed trial (TRL 4)	To verify at TRL 4	R9
PR-010	UR-008, UR-009	Physical architecture	Installation, mass and wind	Two trained people shall fit it in 45 minutes with no drilling or pole wiring on poles of 60 to 140 mm; it shall weigh 3.5 kg or less (relaxed from 3 kg, NSM-DDR-002) and withstand 35 m/s gusts.	45 min; 60 to 140 mm; 3.5 kg; 35 m/ s	max; range; max; min	min; mm; kg; m/s	Analysis (NSM-CAL-001 H, TRL 3); timed trial and weighing (TRL 4)	Met on paper on mass (3.45 kg, margin 0.05 kg), wind and fit; install time to verify at TRL 4	R10
PR-011	UR-011	Software	Contaminated data	Intervals with wind above about 5 m/s or	5	about	m/s	Field comparison with a weather	To verify at TRL 4: method chosen,	R11

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				heavy rain shall be flagged: wind by a level-based flag, rain on the server from weather data (NSM-DDR-002).				station (TRL 4)	threshold needs field data	
PR-012	UR-013	Electrical	Radio use	Radio use shall stay within the EU868 1 % duty cycle and The Things Network fair use of 30 s uplink a day, with a 15 minute interval at SF7 to SF9 lengthened automatically at SF10 to SF12.	30; 1	max	s/day; %	Analysis (NSM-CAL-001 F, TRL 3)	Met on paper: 23.7 s/day at SF9; 29.97 s/day at SF12, with almost no margin for join, time requests and retries	R12
PR-013	UR-014	Physical architecture	Parts cost	The NoiseMap parts (BOM lines 7 to 14) shall cost at or under the value-engineering target; the FieldNode core is costed in the FieldNode repo and the full node cost is stated.	100	max	USD	Priced BOM (NSM-CAL-001 I, TRL 3)	Met on paper: USD 72.00, USD 28.00 under the target; full node USD 211.00	R13
PR-014	UR-010	Materials	Service life	The node shall last 5 years outdoors	5	min	years	Design review (TRL 3); ageing test (TRL 4)	To verify at TRL 4	R14

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				with a yearly windscreen change and one battery change.						
PR-015	UR-015	Software	Openness	Hardware, firmware, calibration method and data format shall be published under the repo licenses.	CERN-OHL-S-2.0; MIT	exact	n/a	Design review (TRL 3)	Met by design	R15
PR-016	UR-016	Safety	Public notice	Each node shall be installed with a notice plate on the pole saying what is measured (levels only) and linking to this repository (NSM-DDR-001 D8).	1 plate per pole	min	plate	Design review (TRL 3); inspection at installation (TRL 4)	To verify at TRL 4: the plate is part of the installation, not of the BOM	new
PR-017	UR-009	Safety	Falling-part retention	A safety lanyard of 2 mm stainless wire rope shall hold the microphone arm to the pole if the arm clamp slips.	1 lanyard	min	lanyard	Design review (build plan Step 11, TRL 3); pull test at TRL 4	Met by design (modelled in cad/src/model.py, BOM line 13)	new

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

Table 2. Assumptions behind the targets.

Assumption	Value	Source or basis
Microphone height	about 4 m above the ground	4.0 ± 0.2 m assessment height for EU strategic noise maps (Directive 2002/49/EC, Annex I)
Quietest night level to resolve	about 35 dBA	Quiet residential street at night (estimate)
Loudest level to resolve	about 110 dBA	Close pass of a loud motorcycle or siren (estimate)
Peak sun hours, winter design case	1.5 h	Mid-latitude winter (estimate)
Wind speed that spoils readings with a foam windscreen	about 5 m/s	Common practice (estimate, to be checked)
Microphone data	ICS-43434: 65 dB SNR, 120 dB SPL overload, 60 Hz to 20 kHz; IM72D128: 72 dB(A) SNR	ICS-43434 product page (TDK InvenSense); IM72D128 from the DNMS README (DNMS)
Street pole diameter	60 to 140 mm, design case 114.3 mm	Common lighting and signal poles (estimate)

Safety: Requirements R8 and R10 involve work at height beside traffic and a lithium cell. See the safety section of NSM-PRC-001.